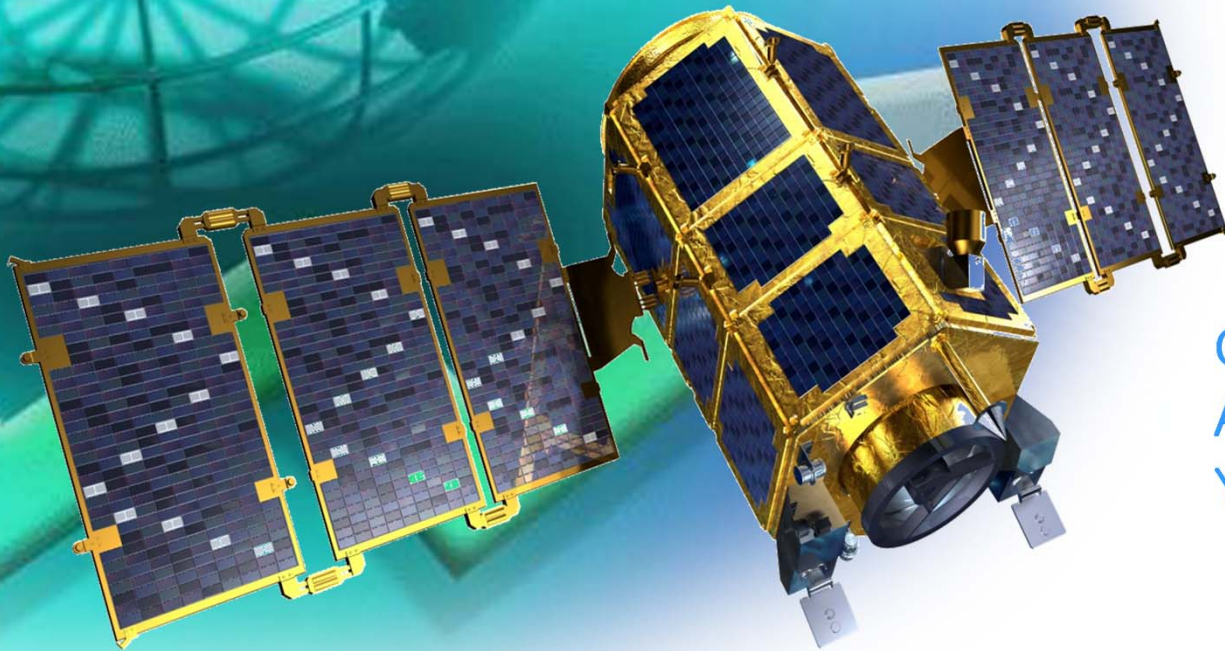
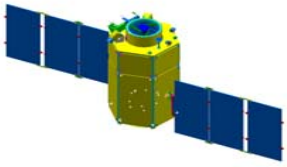


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Spacecraft Systems



Chandeok Park
Astrodynamics & Control Lab.
Yonsei University



Chapter 10

SPACECRAFT DESIGN & SIZING

10.1 Requirements, Constraints, and the Design Process

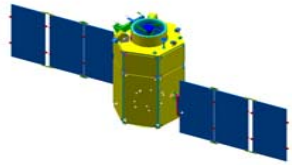
10.2 Spacecraft Configuration

10.3 Design Budgets

10.4 Designing the Spacecraft Bus

10.5 Integrating the Spacecraft Design

10.6 Examples

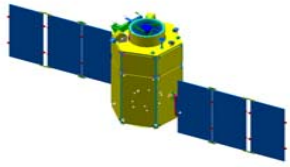


10.5 Integrating the S/C Design

◆ Overview of S/C Design and Sizing (Table 10–1)

TABLE 10-1. Overview of Spacecraft Design and Sizing. The process is highly iterative, normally requiring several cycles through the table even for preliminary designs.

Step	References
1. Prepare list of design requirements and constraints	Sec. 10.1
2. Select preliminary spacecraft design approach and overall configuration based on the above list	Sec. 10.2
3. Establish budgets for spacecraft propellant, power, and weight	Sec. 10.3
4. Develop preliminary subsystem designs	Sec. 10.4
5. Develop baseline spacecraft configuration	Secs. 10.4,10.5
6. Iterate, negotiate, and update requirements, constraints, design budgets	Steps 1 to 5



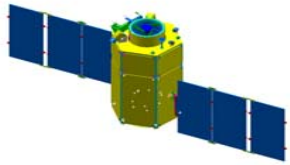
10.5 Integrating the S/C Design

◆ 10.5.1 S/C Size

- Knowledge on the S/C weight and power enables us to estimate its size

TABLE 10-28. Rules for Estimating Volume, Dimension, Area, and Moments of Inertia.
 $M \equiv$ spacecraft loaded mass in kg as defined in Table 10-10.

Characteristic	Estimate	Range (Coefficients)
Volume (m^3)	$V = 0.01 M$	0.005 to 0.05
Linear Dimension (m)	$s = 0.25 M^{1/3}$	0.15 to 0.30
Body Area (m^2)	$A_b = s^2$	—
Moment of Inertia ($\text{kg} \cdot \text{m}^2$)	$I = 0.01 M^{5/3}$	—

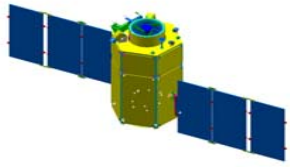


10.5 Integrating the S/C Design

- External solar arrays greatly increase S/C Moment of Inertia, particularly about the axes perpendicular to the array axis
 - Increase in Moment of Inertia $\sim L_a^2 M_a$
 - L_a : length from S/C body center to center of each panel
 - M_a : solar array weight

TABLE 10-29. Rules for Estimating Area Offset and Moment of Inertia of a Solar Array.
 These should be added to the body values computed in Table 10-28. See text for definition of terms.

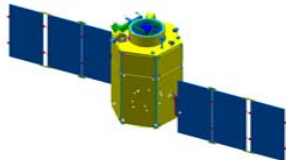
Solar Array Area Offset (m)	$L_a = 1.5 s + 0.5 (A_a / 2)^{1/2}$
Solar Array Moment of Inertia (kg ? m ²)	
Perpendicular to Array Face	$I_{ax} = (L_a^2 + A_a / 12) M_a$
Perpendicular to Array Axis	$I_{ay} = (L_a^2 + A_a / 24) M_a$
About Array Axis	$I_{aa} = (A_a / 24) M_a$



10.5 Integrating the S/C Design

◆ 10.5.2 Lifetime & Reliability

- Reliability is a parameter under the designer's control
 - Should identify the ways in which the S/C may fail
 - Should tailor the design to eliminate or limit failures to acceptable levels
 - Understanding the functions needed for a successful mission enables us to understand the factors which limit mission life or threaten success
- Searching for and identifying the ways in which equipment can fail is a basic part of design for reliability
⇒ Failure Modes Effects and Critical Analysis (FMECA)
- System fail at high rates: Infant Mortality, Wear-out.

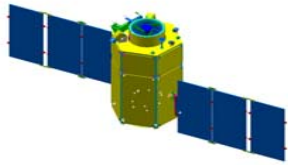


10.6 Examples

◆ FireSat S/C Design

TABLE 10-30. Preliminary Estimate of FireSat Spacecraft Parameters. See text for discussion. These parameters are based primarily on the payload parameters defined in Sec. 9.6.

Parameter	FireSat Estimate	Notes and References
Payload:		
Mass	28 kg	Table 9-15
Power	32 W	Table 9-15
Spacecraft:		
Dry mass	140 kg	Payload mass/0.2; Text + Table 10-5
Average power	110 W	Payload power/0.3; Text + Table 10-9
Solar array power	170 W	Eclipse allowance, Eq. (11-1) ✕
Solar array design	Body-mounted omni array, 1.7 m ² facing the Sun (8.5 m ² total area) ✕	Sec. 10.4.6, array on 5 non-nadir faces
Control approach	3-axis, nadir pointed	Sec. 10.4.2
Propellant:		
ΔV	28 kg	Table 7-3; Eq. (17-7)
Attitude control + residual	2 kg	7%; Secs. 10.4.2, 11.1
Margin	4 kg	15%; Table 10-7
Total propellant	34 kg	Sum of the above
Propulsion approach	Metered bipropellant ($I_{sp} = 300$ s); no kick stage	Text
Spacecraft Loaded Mass:	175 kg	Dry mass + propellant
Spacecraft Size and Moments:		
Volume	1.7 m ³	Table 10-28
Linear dimensions	1.4 m	Table 10-28
Body cross-sectional area	2.0 m ²	Table 10-28
Moment of inertia	50 kg · m ²	Table 10-28



10.6 Examples

◆ FireSat S/C Design (cont'd)

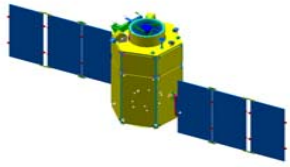
- Two ways of developing a preliminary weight budget

TABLE 10-31. Preliminary FireSat Spacecraft Weight Budget.

Element of Weight Budget	(1)* Est. % of Spacecraft Dry Mass	(2)† Est. % of Payload Mass	(3) Est. Mass Based on Col. (1) (kg)	(4) Est. Mass Based on Col. (2) (kg)	Comments
<i>Payload</i>	20.0	100.0	28.0	28.0	Payload mass estimate from Table 10-30
<i>Structures</i>	21.0	75.0	29.4	21.0	
<i>Thermal</i>	4.5	16.1	6.3	4.5	
<i>Power</i>	30.0	107.1	42.0	30.0	
<i>TT&C</i>	4.5	16.1	6.3	4.5	
<i>Att. Control</i>	6.0	21.4	8.4	6.0	
<i>Prop (dry)</i>	6.0	21.4	8.4	6.0	
<i>Margin (kg)</i>	—	—	11.2	40.0	Note that using the approach in Col. (4) the margin is maintained at the system level, not the subsystem level
<i>Spacecraft Dry Mass (kg)</i>	—	—	140.0	140.0	Estimate from Table 10-30
<i>Propellant Mass (kg)</i>	—	—	35.0	35.0	Estimate from Table 10-30
<i>Spacecraft Loaded Mass (kg)</i>	—	—	175.0	175.0	Sum of spacecraft dry mass & propellant mass
<i>Margin as % of Dry Mass</i>	—	—	8.0%	28.6%	Margin / (Spacecraft dry mass) × 100%

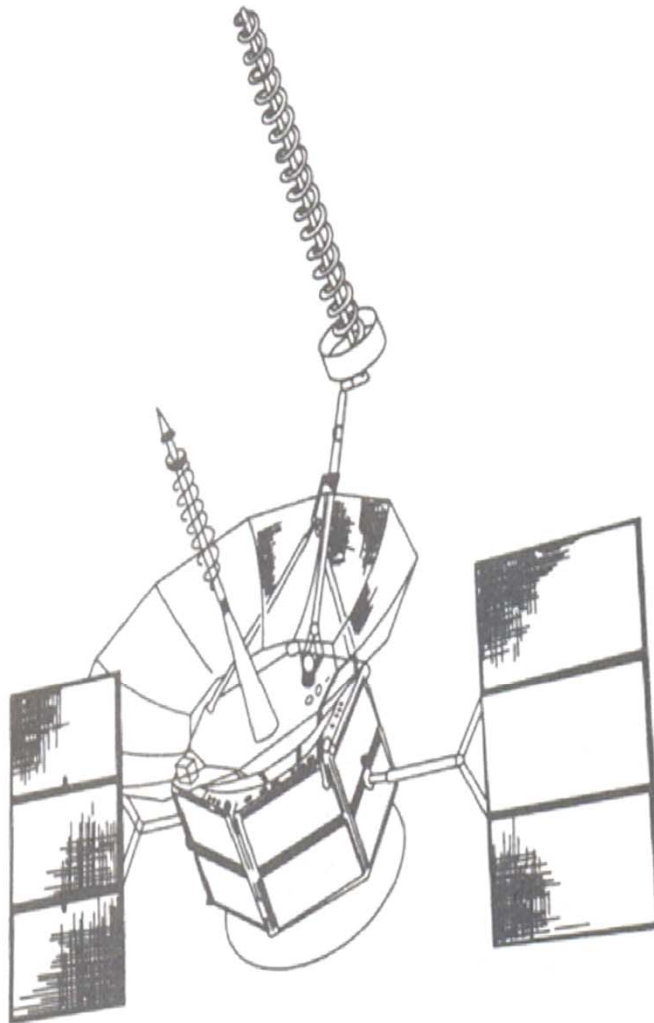
* The percentages in Column (1) are the average values listed in Appendix A.

† The percentages in Column (2) are the average payload values listed in Appendix A.



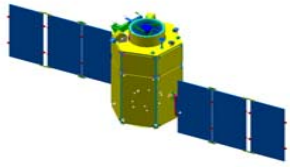
10.6 Examples

◆ FLTSATCOM



FLTSATCOM	
<i>Mission</i>	Military communications
<i>Payload</i>	Communication transponders and antennas
<i>Size</i>	4.9 m transmit antenna 3.10 m helical receive antenna 2.4 m hexagonal × 1 m deep electronics compartment
<i>Wt. (loaded)</i>	927 kg
<i>Power</i>	1,224 W
<i>Pointing</i>	0.25 deg accuracy
<i>Data Rate (selectable)</i>	1 kbps/250 bps
<i>Orbit</i>	Geosynchronous, 2.5 deg inclined
<i>Reliability</i>	0.267 at 5 yrs

Fig. 10-3. Mission Parameters for FLTSATCOM and mass in kg; this use continues throughout the



10.6 Examples

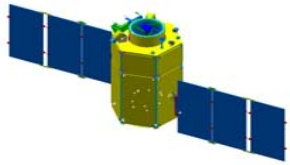
◆ FLTSATCOM (cont'd)

TABLE 10-32. FLTSATCOM

Requirements and Constraints (See Table 10-3).

Space-craft	Mission	Payload	Orbit	Environ-ment	Launch System	Ground System
FLTSAT-COM	Global UHF commu-nications	Communications transponder & antennas	24-hr circular inclined 2.5 deg	Atlas-Centaur launch; 5 yrs in GEO	Atlas-Centaur with solid apogee kick motor	Air Force Satellite Control Network (SGLS)

- UHF: Ultra High Frequency

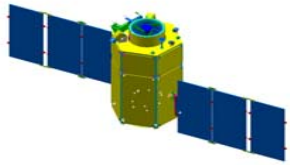


10.6 Examples

◆ FLTSATCOM (cont'd)

TABLE 10-33. Initial Design Decisions for FLTSATCOM

Design Aspect	FLTSATCOM Design Approach
<i>Spacecraft Weight and Power</i>	Dominated by heavy weight and high-power payload; constrained by booster capability
<i>Spacecraft Size</i>	Folded configuration limited by fairing; deployed configuration dominated by antenna size and solar array area
<i>Attitude Control Approach</i>	Spin stabilization for kick motor firing; 3-axis control on orbit
<i>Solar Array Approach</i>	Planar-oriented array
<i>Kick Stage Use</i>	Solid AKM used for orbit insertion
<i>Propulsion Approach</i>	Liquid hydrazine thrusters used for attitude
<i>Orbit Control</i>	Liquid hydrazine thrusters used for attitude control

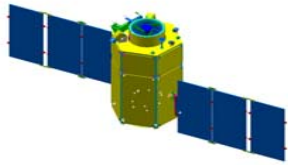


10.6 Examples

◆ FLTSATCOM (cont'd)

TABLE 10-34. Propellant Budget for FLTSATCOM (See Table 10- 7)

Element	Mass (kg)	Design Characteristics
<i>Velocity Correction and Control</i>		
Orbit insertion (AKM)	855.0	$\Delta V = 1,748.8 \text{ m/s}$; $I_{sp} = 285.5 \text{ s}$
Guidance error correction	4.1	9.0 m/s ; $I_{sp} = 215.2 \text{ s}$; $M_{S/C} = 988 \text{ kg}$
Stationkeeping	6.5	14.2 m/s
Station change	52.7	115.2 m/s
<i>Attitude Control</i>		
Spinup and despin	6.0	Spin speed = 60 rpm; S/C inertia = $995 \text{ kg} \cdot \text{m}^2$
Maneuvering while spinning	3.3	Maneuver angle = 65 deg; Inertia = $917 \text{ kg} \cdot \text{m}^2$
Attitude control	8.4	$I_{sp} = 100 \text{ s}$; lever arm = 1.25 m
<i>Residual</i>	1.5	
<i>Total Propellant</i>		
Solid	855.0	
Liquid	82.5	



10.6 Examples

◆ FLTSATCOM (cont'd)

TABLE 10-35. Power Budgets for FLTSATCOM
10-9)

Subsystem	FLTSATCOM Power Budget (W)
<i>Payload</i>	1,070*
<i>Spacecraft Bus</i>	
<i>Propulsion</i>	40
<i>Attitude Control</i>	33†
<i>Communications</i>	9‡
<i>Data Handling</i>	—
<i>Thermal</i>	17
<i>Electric Power</i>	56
<i>Operating Power</i>	1,224
<i>Orbital Average Power</i>	—
<i>Battery Recharge Power</i>	167
<i>End-of-Life Power</i>	1,391
<i>Beginning-of-Life Power</i>	1,800

* Includes 192 W power conversion losses.

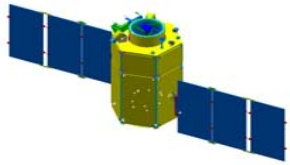
† Includes guidance and navigation functions.

‡ Combined communication and command and data handling.

TABLE 10-36. Weight Budgets for HEAO-B and FLTSATCOM (see Table 10-10)

Subsystem	FLTSATCOM Weight Budget (kg)
<i>Payload</i>	222.0
<i>Spacecraft Bus</i>	
Propulsion	29.5
Attitude Control	57.7
Communications	26.3*
Command and data handling	—
Thermal	14.5
Electric power	336.0
Structure	154.0
<i>Spacecraft Dry Weight</i>	841.0
<i>Propellant</i>	85.4†
<i>Ballast</i>	—
<i>Loaded Weight</i>	927.0
<i>Apogee Kick Motor</i>	916.0
<i>Injected Weight</i>	1844.0
<i>Adapter</i>	19.5
<i>Boosted Weight</i>	1863.0

* Combined communication and command and data handling. † Includes 2.8 kg pre

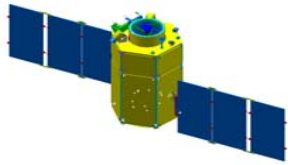


10.6 Examples

◆ FLTSATCOM (cont'd)

TABLE 10-37. Characteristics of FLTSATCOM Propulsion SubSystem

Characteristic	FLTSATCOM
<i>Propellant</i>	Hydrazine monoprop.
Capacity	83 kg
Tankage	Two 0.6 m diameter titanium
Pressurant	2.8 kg nitrogen
Tank weight	14.4 kg
<i>Lines and Valves</i>	Central isolation valve distribution assembly
Weight	7.5 kg
<i>Thrusters</i>	Sixteen 4.4 N (4 roll, 4 pitch, 8 yaw and ΔV) Four 0.44 N (roll-yaw)
Power Consumption	40 W (catalyst bed and line heaters)
Weight	7.6 kg

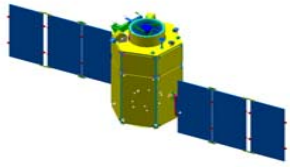


10.6 Examples

◆ FLTSATCOM (cont'd)

b **TABLE 10-38. Attitude-Control Requirements for FLTSATCOM** (Table 10-13)

Requirement	FLTSATCOM
<i>Payload Requirements</i>	
Article to be pointed	Communication antennas
Pointing direction	Nadir
Pointing accuracy	0.25 deg
Pointing stability	—
Slew rate	—
Exclusion	—
<i>Other Requirements</i>	
Sun pointing	Yaw (Z-axis) controlled to keep Y-axis normal to orbital plane
Pointing during ΔV	Yaw (Z-axis) control as above

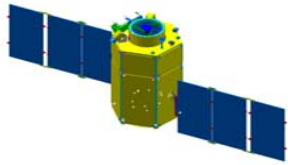


10.6 Examples

◆ FLTSATCOM (cont'd)

TABLE 10-39. Weight and Power for the Attitude-Control Component

Component	FLTSATCOM	
	Mass (kg)	Power (W)
<i>Sensors</i>		
Sun sensors	1.5	0
Earth sensors	7.2	12
Gyroscopes	—	—
<i>Processing Electronics</i>	23.8	12.8
<i>Reaction Wheels</i>	11.3 (2)	8
<i>Solar Array Drives</i>	13.9 (2)	0.2 (2)
<i>Total</i>	57.7	33

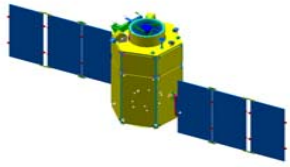


10.6 Examples

◆ FLTSATCOM (cont'd)

TABLE 10-41. Characteristics and Components of the Electric Power SubSystem

Characteristics and Components	FLTSATCOM
<i>Solar Arrays</i> Power output BOL Size Weight	1,800 W BOL normal incidence 1,200 W @ 7 year 23 deg incidence Two 2.8 m × 3.8 m panels 92.6 kg
<i>Battery</i> Capacity Weight	600 W · hr (24 A · hr) 89.6 kg total
<i>Power Control</i> Type Weight Power	Unregulated array 9.7 kg 12.9 W
<i>Power Switching</i> Type Weight Power	Central switching assemblies (1 for S/C, 2 for P/L) 29.3 kg 4.3 W (S/C bus)
<i>Power conversion</i> Type Weight Power	Central S/C bus converter; Separate P/L converters 8.5 kg S/C bus, 78.3 kg P/L 14.3 W S/C bus, 192 W P/L
<i>Cabling</i> Weight Power	72.7 kg 24 W



10.6 Examples

◆ FLTSATCOM (cont'd)

TABLE 10-42. Structural Characteristics of FLTSATCOM

Characteristic	FLTSATCOM
Weight	154 kg
Design Approach	
Primary structure	Central cylinder
Solar array	Aluminum honeycomb panels deployed in two wings
Equipment compartment	Honeycomb panels mounted to longeron-stringer frame
Deployables	Wrapped around or folded over equip compartment during boost

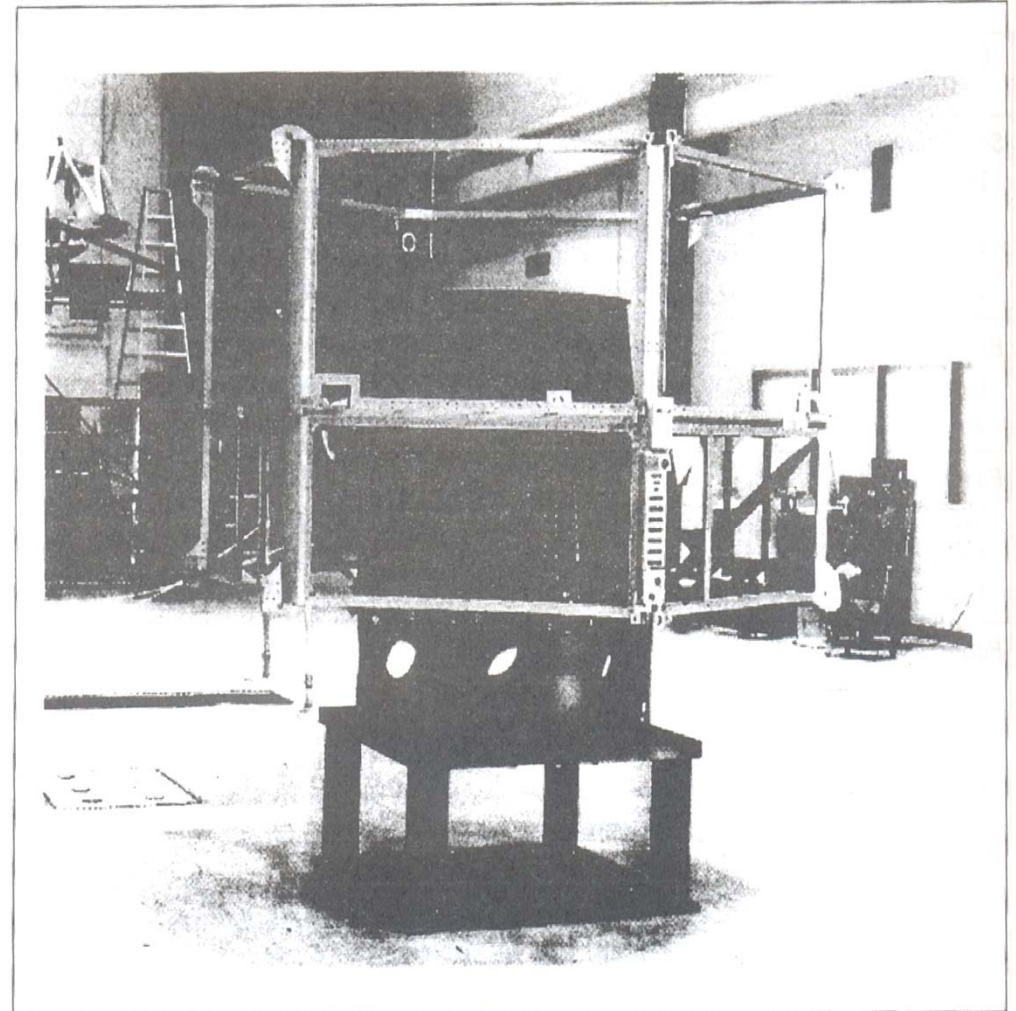
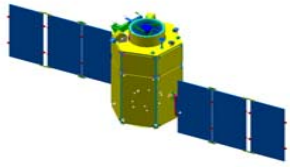


Fig. 10-4. FLTSATCOM Structure.



10.6 Examples

◆ FLTSATCOM (cont'd)

TABLE 10-43. Estimated vs. Actual Sizes of FLTSATCOM and I

Spacecraft Parameters	Units	FLTSATCOM	
		Estimated	Actual
Spacecraft Loaded Weight	kg		927
Spacecraft Power BOL	W		1800
Volume		9.27	
Compartment	m ³		5.0
Folded Spacecraft	m ³		28.2
Linear Dimension			
Diameter	m	2.9	2.4
Length	m		5.1
Body Area	m ²	8.5	12.3
Solar-Array Area	m ²	18	21.5
Offset to Solar-Array Area	m	5.9	5.2
Moments of Inertia			
Folded w/ AKM	kg · m ²	881.3	916.4
w/o AKM			828.5
Deployed I _{xx}		3453.3	3388.9
I _{yy}		935.3	825.8